Biomedical Signal Processing and Control 113 (2026) 108840

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

A review of facial and gait analysis for auxiliary stroke care: Computer vision-based solutions☆

Xiao Han [a,d], Li-Ping Li [b], Zi-Yan Wang [a,d], An-Ying Xu [c], Tian-Shu Wang [d], Tao Yang [d], Kong-Fa Hu [a,d,*]

[a] Institute of Traditional Chinese Medicine Literature, Nanjing University of Chinese Medicine, No. 138, Xianlin Road, Nanjing, 210023, China
[b] Department of Rehabilitation Medicine, The First Affiliated Hospital of Nanjing Medical University, No. 300, Guangzhou Road, Nanjing, 210029, China
[c] School of Mechanical Engineering, Guizhou University, TianyuanNorthRd, Guiyang, 550025, China
[d] College of Artificial Intelligence and Information Technology, Nanjing University of Chinese Medicine, No. 138, Xianlin Road, Nanjing, 210023, China

ARTICLE INFO

Keywords:
Stroke detection
Stroke rehabilitation
Facial analysis
Gait analysis
Deep learning
Machine learning

ABSTRACT

Stroke remains a principal contributor to global mortality and long-term disability, with early diagnosis and continuous rehabilitation being critical for patient outcomes. Facial asymmetry and gait impairment are key clinical indicators; however, their assessment often relies on subjective scales. This review is the first to systematically explore computer vision-based solutions for stroke care, with a focus on automated facial and gait analysis. We evaluate a wide range of approaches for both facial and gait analyses. These include traditional machine learning and deep learning architectures for facial feature extraction, alongside skeleton keypoint kinematics, optical flow, and multimodal fusion strategies for gait characterization. Our analysis reveals that vision-based technologies can generate objective, quantitative biomarkers, offering significant potential for enhancing early stroke detection, severity classification, and quantitative rehabilitation monitoring. However, challenges such as sensitivity to imaging conditions, data scarcity, and limited clinical interpretability remain barriers to widespread adoption. This review determines that vision-based solutions, particularly those leveraging multimodal data fusion, offer a promising alternative to traditional assessment methods and wearable sensors. These solutions pave the way for more accessible, efficient, and personalized tele-rehabilitation scenarios, thereby bridging the gap between technological innovation and clinical integration.

Contents

1. Introduction 2
2. Vision-based facial analysis 3
   2.1. Traditional machine learning approaches 3
   2.2. Deep learning approaches 4
      2.2.1. Convolutional neural network 4
      2.2.2. Sequence-based neural network 5
   2.3. Comparative analysis of traditional machine learning and deep learning approaches 5
3. Vision-based gait analysis 5
   3.1. Skeletal keypoint-based features 6
   3.2. Optical flow-based features 8
   3.3. Fusion-based features 9
4. Discussion 9
   4.1. Technical and clinical limitations 9
   4.2. Bridging vision-based technologies and clinical care 10
   4.3. Vision-based alternatives to wearable sensor challenges 10

☆ This document is funded by National Key Research and Development Program of China (Grant No. 2022YFC3502302)

* Corresponding author.
E-mail addresses: x.han@njucm.edu.cn (X. Han), 2319689603@qq.com (L.-P. Li), zywang216@njucm.edu.cn (Z.-Y. Wang), gs.ayxu@gzu.edu.cn (A.-Y. Xu), wangtianshu@njucm.edu.cn (T.-S. Wang), yangtao@njucm.edu.cn (T. Yang), kfhu@njucm.edu.cn (K.-F. Hu).

https://doi.org/10.1016/j.bspc.2025.108840
Received 12 March 2025; Received in revised form 22 September 2025; Accepted 5 October 2025
Available online 14 October 2025
1746-8094/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

4.4. Future directions 11
5. Conclusion 11
CRediT authorship contribution statement 11
Declaration of competing interest 11
Acknowledgments 11
Data availability 11
References 12

1. Introduction

A stroke is a condition caused by the blockage or rupture of blood vessels in the brain, which impairs cerebral tissue and leads to functional or structural impairments [1]. Strokes are broadly categorized into two types: ischemic and hemorrhagic [2]. Ischemic strokes result from occlusion of vessels supplying the affected region, causing insufficient delivery of nutrients and oxygen, whereas hemorrhagic strokes arise from vessel rupture, causing increased pressure on surrounding tissues and nerves. Globally, approximately 13.7 million people experience stroke each year, with a trend toward younger onset ages [3]. It is estimated that approximately one quarter of individuals over the age of 25 will experience a stroke in their lifetime. Stroke is not only the primary cause of disability worldwide but also the second leading cause of death globally [4]. Current clinical guidelines recommend the use of medical imaging for detection and assessment, employing computer vision technology to identify ischemic or hemorrhagic lesions in the brain, including infarcts and hematomas. These imaging features correspond to the pathophysiological changes associated with stroke, with computed tomography (CT) and magnetic resonance imaging (MRI) remaining the diagnostic gold standards [5,6].

Although medical imaging provides the gold standard for diagnosing intrinsic stroke pathology, clinical manifestations remain crucial for supporting diagnostic decisions and assessing a patient's condition. Standardized assessment tools in clinical practice enable rapid diagnosis and severity evaluation on the basis of functional performance, including the National Institutes of Health Stroke Scale (NIHSS), the Cincinnati Prehospital Stroke Scale (CPSS) [7], and the Face Arm Speech Test (FAST) [8], which are widely used for swift diagnosis and quantification of stroke severity [9]. The expanded Balance Eye Face Arm Speech Test (BE-FAST), which incorporates balance and visual assessments, further improves diagnostic accuracy, reducing misdiagnosis rates to 9.9% [10]. Face and gait, as key observational indicators within these scales, are closely associated with functional and structural damage in post-stroke brain regions [11,12], indicating significant diagnostic and prognostic value in the evaluation of stroke patients. Advances in computer vision technology help address the lack of specificity in standardized scales for facial and gait assessment [13] while reducing the influence of evaluator subjectivity. This demonstrates considerable potential for enhancing assessment accuracy, serving as an effective auxiliary tool and offering new perspectives for stroke diagnosis and functional evaluation.

Facial paralysis can present as unilateral or bilateral loss of facial muscle motor function, often accompanied by drooping facial features, and is a classic manifestation of stroke [14,15]. This dysfunction not only compromises basic physiological functions such as speech, eye closure, swallowing, and eating but also directly affects an individual's ability to communicate nonverbally by impeding natural facial expressions, thereby providing clinicians with important information regarding the patient's condition [16]. Approximately 30% of stroke patients experience acute central facial paralysis, which can serve as an early warning sign [17]. However, central facial paralysis is frequently confounded with peripheral facial nerve palsy. Accurate differentiation is crucial for timely clinical intervention, significantly reducing the risk of treatment delays due to misdiagnosis during triage. Vision-based approaches, which encompass machine learning algorithms and neural network models, effectively extract multiple clinically relevant features, such as facial landmark displacement, joint angle changes, and dynamic motion trajectories. Diagnosis based on facial features can accelerate initial disease screening and monitoring, enabling earlier initiation of comprehensive treatment [18]. Moreover, quantitative analysis of facial dysfunction, including symmetry, range of motion, and coordination patterns, allows clinicians to objectively assess neurological impairment severity and monitor rehabilitation progress [19].

Gait impairment is another common and critical clinical manifestation following stroke. Over 60% of stroke survivors are unable to walk independently after onset [20], and 83% experience balance disorders [21]. Even after rehabilitation therapy, more than half of patients still experience gait disorders, severely compromising their quality of life [22]. These findings underscore the critical importance of monitoring and evaluating post-stroke gait function recovery. In the context of advancing gait recognition technology, capabilities in movement pattern analysis have significantly improved. Currently, gait analysis and assessment technologies primarily fall into two categories: wearable sensor–based technologies and vision-based technologies. Wearable sensor technology employs miniature devices attached to the body surface, such as inertial measurement units (IMUs) [23], accelerometers [24], and pressure sensors [25], to collect real-time physiological signals and motion state data related to gait. Although wearable devices enable real-time motion data collection, they are relatively expensive, require lengthy single-session gait analysis, and demand high operator expertise. This poses limitations for long-term assessment in stroke patients, particularly those with limb dysfunction or cognitive impairments. In contrast, vision-based technology, with its noncontact and low-interference characteristics, better aligns with the clinical demands of gait analysis for stroke patients and has become a current research hotspot. This technology captures two-dimensional features of human motion via optical cameras (e.g., standard cameras, smartphones) or acquires three-dimensional motion information via depth sensors (e.g., Kinect, RealSense), combining computer vision algorithms to track and analyze movement behaviors. In recent years, rapid advancements in computer vision have driven significant progress in vision-based gait analysis. A series of efficient and robust pose estimation algorithms (e.g., [26–28]), optical flow analysis methods (e.g., [29,30]), and multimodal fusion strategies (e.g., [31,32]) has been successfully applied to extract gait features and identify abnormal patterns in stroke patients. These algorithms not only increase the accuracy and efficiency of motion tracking but also effectively capture stroke-related specific gait parameters, such as gait asymmetry, spatiotemporal parameter variability, and abnormal joint range of motion. This provides more objective and refined quantitative evidence for clinical assessment and rehabilitation intervention.

This review explores the application of computer vision in stroke assisted diagnosis and rehabilitation assessment, focusing on two core areas: facial and gait analysis. The specific research objectives include the following:

1. To summarize and compare facial analysis methods based on traditional machine learning and deep learning approaches, analyzing differences in feature representation capabilities and clinical applicability.

2. To evaluate the effectiveness and applications of vision-based gait analysis techniques for quantifying stroke-related gait impairments through methods such as skeleton keypoint-based kinematic parameter extraction, optical flow dynamic analysis, and multimodal fusion strategies.
3. To identify the technical gaps and limitations encountered during the clinical translation of existing technologies.

This paper is structured as follows: Section 2 introduces vision-based facial analysis methods, including traditional machine learning algorithms and deep learning models. Section 3 examines vision-based gait analysis techniques, focusing on kinematic parameter calculation via skeletal landmarks, dynamic feature analysis of optical flow, and the effectiveness of multimodal fusion methods in complex scenarios. Section 4 analyzes the limitations and challenges of existing approaches and proposes future directions for multimodal fusion and clinical applications. Section 5 concludes by summarizing the strengths and suitable application contexts of vision-based facial recognition and gait analysis technologies in stroke diagnosis and rehabilitation.

2. Vision-based facial analysis

Clinical facial signs, such as facial drooping, mouth and eye deviations, and slurred speech, are key diagnostic indicators for stroke diagnosis and rehabilitation monitoring [18]. Facial weakness, typically caused by brain or facial nerve damage, manifests as diminished control over facial muscles and may progress to paralysis, a common clinical symptom of stroke [33]. As one of the most prevalent facial nerve dysfunctions, facial paralysis is characterized primarily by muscle drooping or reduced strength, often leading to difficulties in chewing, speech, swallowing, and emotional expression [34]. Furthermore, facial morphology and motor function play crucial roles not only in aesthetics and emotional expression, but also in corresponding facial changes, which serve as quantitative biomarkers for assessing stroke severity and rehabilitation outcomes [35]. Commonly used clinical tools for facial paralysis evaluation include the House–Brackmann grading system (HB) [36], the Toronto facial grading system [37], the facial nerve grading system 2.0 (FNGS 2.0) [38] and the Sunnybrook facial grading system [39]. These systems typically require patients to perform a series of specified movements. On the basis of clinical observations and the clinician's subjective judgment, facial nerve function is classified into different grades, generally ranging from Grade I (normal function) to Grade VI (complete absence of movement). Although these assessment systems provide an important standardized framework for quantifying facial nerve function, their application relies on the subjective judgment of clinicians, which presents certain limitations. To overcome this limitation, researchers have begun exploring facial image analysis methods based on machine learning and neural networks, aiming to achieve objective and rapid assessment of paralysis severity. This section systematically evaluates computational methods for facial analysis in stroke assessment, focusing on three clinical indicators: facial weakness, facial paralysis, and abnormal facial expressions.

2.1. Traditional machine learning approaches

Facial weakness is a common stroke symptom frequently associated with impaired facial muscle control resulting from nerve damage [34, 40]. Zhuang et al. [41] developed a facial video classification system that uses a histogram of oriented gradients (HoG) features [42] in combination with penalized linear discriminant analysis (pLDA) [43]. The system categorizes input videos into three classes: normal, left-sided weakness, and right-sided weakness. HoG features are robust to local misalignments and effectively capture subtle gradient changes associated with facial weakness, enabling low computational complexity while maintaining robust performance against local deformations [42]. In the following year, the team integrated HoG features with facial landmarks, achieving a significant improvement in classification accuracy for facial weakness [34]. Similarly, Aldridge et al. [40] employed a parallel approach that combines HoG with facial landmark detection to compare human observers and machine learning algorithms in identifying facial weakness. Their results demonstrated that machine learning algorithms outperformed human observers in determining the hemiplegic side of facial paralysis in stroke patients. These finding suggest that computer vision algorithms can assist in the clinical assessment of facial weakness, providing crucial support for early diagnosis and objective evaluation.

Facial paralysis refers to dysfunction of facial muscles caused by abnormal nerve innervation and clinically manifests as asymmetry in facial appearance and movement [33]. It is traditionally classified into peripheral and central types. Central facial paralysis, commonly caused by stroke due to damage to the unilateral motor cortex, predominantly affects the lower half of the face [44]. Although facial paralysis itself is not always directly caused by stroke, its occurrence can serve as an important precursor to stroke [45]. Therefore, achieving early intervention and accurate diagnosis is crucial for improving patient prognosis and promoting functional recovery [46]. In RGB image-based facial paralysis detection, mainstream approaches construct facial configuration features by extracting key facial landmarks, such as calculating distances and angles between points and classifying them on the basis of these geometric characteristics. Gaber et al. [47] developed a Kinect-based system that quantifies facial paralysis by calculating symmetry indices for the eyebrow, eye, and mouth regions. The team [46] further utilized the Kinect system to capture the values of different facial animation units and their dynamic variation characteristics. They employed support vector machines (SVMs) [48] to classify facial paralysis severity into seven categories: left/right mild, left/right moderate, left/right severe, and normal. Similarly, Guo et al. [49] employed an SVM to automatically assess facial paralysis severity by extracting features from facial landmarks in images of patients, using the HB grades as the ground truth for training the classifier. Park et al. [50] assessed facial motor dysfunction by calculating asymmetry indices based on displacement differences between the left and right sides of the forehead and mouth regions. Building upon this, Barbosa et al. [51] advanced the quantification of facial symmetry by constructing refined geometric features including bilateral iris area ratios and distance ratios between facial landmarks. Their study also evaluated multiple classifiers, including random forests [52] and decision trees [53], to improve accuracy in distinguishing between peripheral and central facial paralysis. Although the extracted iris features effectively reflect facial symmetry, the method struggles with complex scenarios such as wrinkles, facial hair, or occlusions, limiting its practical application in clinical settings. He et al. [54,55] proposed an evaluation method based on facial motion optical flow information. Their earlier work [54] quantified symmetry in motion intensity and direction across bilateral facial regions through optical flow analysis and employed a radial basis function (RBF) [56] network for regional paralysis grading. However, this approach suffers from high computational demands for optical flow calculations and struggles to ensure accuracy when estimating changes in standardized facial actions (e.g., raising eyebrows, gently closing eyes, squinting, wrinkling the nose, and smiling). To overcome these limitations, [55] proposed a motion feature extraction method based on multiresolution local binary patterns (LBPs). This approach quantifies motion symmetry by calculating the resistor-average distance between LBP features extracted from both sides of the face and employs an SVM for quantitative facial paralysis assessment. Compared with traditional static features, the unique advantage of optical flow in capturing subtle dynamic information of facial muscle movements enables more effective assistance in the quantitative diagnosis and severity evaluation of facial paralysis.

In addition to the use of external geometric features for quantification, assessment methods based on intrinsic physiological signals such as blood flow and temperature also offer new pathways for evaluating

Table 1
Summary of traditional machine learning-based facial feature extraction methods for stroke evaluation, organized by year.

Year	Study	Objective	Facial features
2007	He et al. [54]	Assess paralysis severity	Motion symmetry (optical flow)
2015	Gaber et al. [47]	Quantify facial paralysis	Symmetry indices (eyebrow, eye, mouth)
2015	Park et al. [50]	Assess motor dysfunction	Forehead and mouth symmetry
2015	Liu et al. [57]	Evaluate nerve function	Facial temperature asymmetry
2017	Guo et al. [49]	Assess paralysis severity	Facial landmarks
2019	Zhuang et al. [41]	Classify facial weakness	HoG, pLDA
2019	Barbosa et al. [51]	Distinguish peripheral vs. central paralysis	Iris area ratios and facial distance ratios
2020	Zhuang et al. [34]	Classify facial weakness	HoG, facial landmarks
2020	Jiang et al. [58]	Assess paralysis severity	Blood flow distribution
2022	Aldridge et al. [40]	Identify facial weakness	HoG, facial landmarks
2022	Gaber et al. [46]	Classify paralysis severity	Dynamics of facial action units
2022	Xu et al. [55]	Assess facial paralysis	LBP

facial paralysis. Jiang et al. [58] introduced laser speckle contrast imaging technology. By measuring facial blood flow to generate RGB color images and blood flow images, they segmented regions of interest on the patient's face, extracted blood flow distribution characteristics, and achieved quantitative assessment of facial paralysis severity based on the HB facial nerve grading system. Separately, infrared thermography studies revealed markedly greater interside temperature differences in facial paralysis patients than that in healthy individuals, with this asymmetry closely correlated with the extent of facial nerve damage, providing crucial evidence for objective clinical assessment [59]. Liu et al. [57] developed an automated facial nerve function evaluation method based on infrared thermography, that is capable of automatically segmenting facial regions and quantifying temperature asymmetry through comparative analysis. The study compared three classification algorithms: the RBF network, SVM, and k-nearest neighbors (KNN) [60]. The experimental results demonstrated that the RBF network achieved the highest accuracy in facial paralysis classification, indicating the promising application potential of infrared thermography in automated facial paralysis assessment.

Table 1 provides an overview of traditional machine learning-based methods for facial analysis in stroke evaluation, summarizing the objectives and types of facial features used in each study. Studies published prior to 2017 predominantly relied on symmetry metrics derived from facial landmarks to assess paralysis and motor dysfunction. By contrast, research from 2019 onward shows a growing emphasis on HoG features, which are often combined with facial landmarks or used alongside classifiers such as pLDA for improved classification of facial weakness. This shift suggests a trend toward more robust and discriminative feature representations.

2.2. Deep learning approaches

2.2.1. Convolutional neural network

Convolutional neural networks (CNNs) are a type of deep learning algorithm inspired by the biological visual system and are widely applied in image recognition tasks. Processing begins by applying a set of filters to the input image, followed by rectification of the output via a rectified linear unit (ReLU) to generate "feature maps" [61]. Max pooling is then employed to reduce dimensions and enhance feature extraction. After multiple iterations of convolution and pooling, the network terminates with fully connected layers. The number of output layer units matches the number of categories, enabling image classification. Gemma et al. [15] introduced a multilayer perceptron-based method for facial landmark detection by identifying and tracking key facial features (e.g., eyebrows, eyes, nose, and mouth) to quantify asymmetry. This is achieved by comparing the positional and angular differences of these landmarks between the paralyzed and healthy facial sides, yielding geometric metrics for the objective assessment of paralysis severity. Hossain et al. [62] used three deep learning models, namely, ResNet-50 [63], Inception-V3 [64], and VGG-16 [65], to extract and analyze deep features from the Kaggle public dataset "Facial Droop and Facial Paralysis Images", which contains 1,024 facial stroke images. Although all three models demonstrated potential for detecting facial paralysis, ResNet-50 achieved the highest accuracy at 96.63%. Similarly, Alsharif et al. [35] employed transfer learning techniques based on ResNet-50 using the same dataset. By analyzing patients' mouth and facial expressions, they developed a stroke symptom recognition system that effectively enhanced medical diagnostic capabilities for stroke. These two studies [35,62] demonstrate the effectiveness of ResNet-50, where its deep architecture learns more complex high-level semantic feature representations whereas its skip-connection mechanism promotes the reuse of lower-level features, further optimizing the learning process. Despite the diverse sources of publicly available Kaggle datasets, these datasets lack relevant clinical annotation information. Building upon machine learning SVM methods, Guo et al. [66] further explored feature extraction using a fine-tuned GoogLeNet [67] model to predict and assess the severity of unilateral peripheral facial paralysis. Song et al. [33] employed DeepID CNN [68] to automatically extract high-level features from images and introduced the Inception-V3 convolutional neural network to analyze these features, achieving facial paralysis assessment through a comprehensive evaluation of seven facial movements. This approach, which combines Inception-V3 with DeepID, achieved the highest accuracy of 99.3% in identifying severe facial paralysis (L3/R3) whereas for moderate paralysis (L2/R2), the accuracy was slightly lower at 93.3%, and overall, the model demonstrated high clinical reference value. Yolcu et al. [69] developed an automated facial expression recognition system to assist clinicians in detecting neurological disorders. The system employs a four-stage CNN architecture: the first three CNNs perform independent feature extraction and segmentation for the eyebrow, eye, and mouth regions. These three branches operate in parallel, with their outputs collectively forming a compact feature representation termed "face-iconization". This feature, combined with the original facial image, is fed into the fourth CNN for integration and expression classification. Isolated local analysis may yield false positives owing to individual variations (e.g., natural asymmetry) or transient movements (e.g., speaking). This approach incorporates holistic facial image information, providing a reasonable baseline reference for evaluating local features. Facial weakness or paralysis can lead to difficulties in speech, swallowing, and emotional expression, potentially triggering anxiety and depression, significantly impacting the quality of life. Therefore, the ability to automatically recognize emotions through facial expressions could aid in the early detection of patients' psychological stress and emotional changes, thereby providing crucial support for treatment planning. Xu et al. [70] collected 90 video segments from 45 facial paralysis patients (spanning three severity grades), which captured six basic emotional expressions. By employing transfer learning with healthy facial images, their study effectively mitigated the training challenges associated with deep learning models based on the VGG architecture under data-scarce conditions. The improved model achieved 66.58% accuracy in basic emotion recognition, representing a 19.63% increase over the baseline models trained solely on facial paralysis data.

Table 2
Review of deep learning-based facial feature extraction methods for stroke evaluation.

Study	Objective	Architecture	Features
Gemma et al. [15]	Assess paralysis severity	Multilayer perceptron	Geometric features from facial landmarks
Hossain et al. [62]	Detect facial paralysis	ResNet-50, Inception-V3, VGG-16	Deep features
Alsharif et al. [35]	Diagnose stroke	ResNet-50	Deep features from facial expression
Guo et al. [66]	Assess unilateral paralysis severity	GoogLeNet (Inception-v1)	Deep features
Song et al. [33]	Assess facial paralysis	DeepID CNN	Deep features
Yolcu et al. [69]	Detect neurological expressions	Four-stage CNN	Deep features (face-iconization)
Wang et al. [71]	Identify stroke patients	Ensemble (Xception, ResNet, VGG, EfficientNet)	Deep features
Xu et al. [72]	Assess facial paralysis	Dual-path LSTM	Spatiotemporal features
Zhuang et al. [73]	Classify facial paralysis	Bidirectional LSTM	HoG features from landmarks
Xu et al. [70]	Analyze paralysis motion	VGG	Deep features (emotion)

Additionally, central cyanosis (bluish-purple discoloration of the skin or mucous membranes due to decreased blood oxygenation) often manifests in the central facial region. Severe stroke may cooccur with various neuromuscular dysfunctions that affect the face and limbs. Color changes in this area provide crucial clues regarding blood oxygen levels and cerebral blood flow to vital organs [11]. Wang et al. [71] proposed a CNN model that integrates Xception [74], ResNet, VGG, and EfficientNet [75] to identify stroke patients by analyzing cyanosis in the central facial region. Cyanosis not only serves as a potential indicator of stroke but is also commonly observed in various cardiovascular diseases, such as atherosclerosis, hypertension, and heart failure [76]. To increase diagnostic specificity, the study introduced a multimodal fusion strategy that combines facial features with clinical parameters, such as coagulation indicators and white blood cell counts. To quantitatively predict stroke risk, a CNN model, that calculated the corresponding stroke probability based on even subtle dark or bluish-purple changes in the facial center region was employed. This research pioneers new applications of facial analysis in stroke screening and holds significant promise for more efficient and timely clinical interventions.

2.2.2. Sequence-based neural network

Images provide only static information regarding facial features and cannot capture dynamic changes. To effectively identify facial muscle weakness and subtle changes in expressions, combining images with time series information is essential. In this context, recurrent neural networks (RNNs) play a crucial role. RNNs possess a recurrent connection structure that enables them to incorporate historical information from other neurons and from themselves [77]. This configuration allows RNNs to process sequence data of variable length and to model temporal dependencies. However, traditional RNNs are susceptible to gradient explosion or vanishing gradient problems during training, which limits their ability to capture long-term dependencies. Long short-term memory (LSTM) networks address these limitations by introducing a gating mechanism that preserves information over extended periods, making them particularly well suited for time series prediction and classification tasks.

Xu et al. [72] extracted differential features from bilateral facial regions across a sequence of frames in facial videos. These feature sequences were then fed into a dual-path LSTM network to fuse spatiotemporal information for facial paralysis assessment. Zhuang et al. [73] analyzed patient video sequences, extracting HoG feature histograms from 68 key points in the most prominent facial action frames to capture edge and texture information. These were then fed into a bidirectional LSTM to classify facial paralysis based on the shape and appearance features of each target frame. Current research predominantly focuses on automatically identifying the severity of facial muscle weakness, whereas insufficient attention is given to patients' emotional factors.

Table 2 provides a systematic overview of deep learning-based approaches for facial analysis in the context of stroke evaluation, highlighting the diversity in architectural designs and feature representations used across recent studies. A range of network architectures, from classic MLP and VGG to more modern structures such as ResNet, Inception, and GoogleNet, is employed. More recent studies have also incorporated LSTM-based models (e.g., dual-path LSTM and bidirectional LSTM) to capture spatiotemporal features, reflecting a shift toward modeling dynamic facial impairments. Although several studies use deep features learned end-to-end from raw images, some combine these features with handcrafted features such as HoG or geometric features derived from facial landmarks. This hybrid approach suggests an interest in leveraging both learned representations and engineered features for improved performance.

2.3. Comparative analysis of traditional machine learning and deep learning approaches

When facial features, including both static appearance and dynamic movements, are analyzed for tasks such as stroke prediction, traditional machine learning methods and deep learning approaches have distinct advantages and limitations.

Traditional machine learning methods (e.g., SVM and random forests combined with handcrafted features such as HoG) often demonstrate high accuracy on moderately sized datasets, along with computationally efficient training and inference processes. However, a core limitation lies in their heavy reliance on manually designed and annotated features (e.g., facial landmark extraction). This process requires substantial domain expertise and restricts the ability of the model to capture complex static features and subtle dynamic changes. Although feature selection and model decision-making remain relatively interpretable and help gain clinician trust, these methods are still severely constrained by data scale and diversity, exhibiting poor adaptability to real world scenarios involving occlusions or lighting variations.

By contrast, deep learning models (e.g., CNNs and LSTMs) automatically extract multilevel feature representations from raw data through end-to-end learning, bypassing the cumbersome and inherently limited manual feature engineering of traditional approaches. Although they require longer training times and greater computational resources, these models typically achieve higher prediction accuracy and stronger generalization capabilities [78], effectively capturing complex high-dimensional facial patterns [35,62]. For dynamic facial features, analysis relies on capturing facial changes over time, such as subtle muscle movements during speech. Such tasks require not only spatial feature extraction but also temporal modeling, a domain in which recurrent structures such as LSTMs are specifically designed to handle spatiotemporal dynamics. However, CNNs and LSTMs typically demand substantial amounts of high-quality annotated data, and their training processes involve high computational costs. Additionally, the decision-making mechanisms of these models lack interpretability, functioning as "black boxes", which significantly affects clinicians' acceptance and trust in the results during clinical applications.

3. Vision-based gait analysis

Normal gait is the process by which the lower limbs propel the body forward through rhythmic movement, characterized by minimizing

energy expenditure through coordinated movement patterns. Stroke patients struggle to coordinate gait effectively, requiring greater energy to maintain forward motion while also experiencing secondary changes caused by various pathological factors, including primary issues such as decreased muscle strength, neuromuscular control disorders, spasticity, and pain. These subsequently lead to secondary changes such as restricted joint mobility and foot deformities, ultimately manifesting as characteristic gait dysfunction during various phases of the gait cycle. In clinical practice, the severity of gait dysfunction is typically assessed via standardized clinical scales and tests, such as the 6 min walk test [79], the 10-m walk test [80], and the Fugl-Meyer assessment [81]. Although these tools offer rapid application and minimal time requirements, they are susceptible to subjective judgment, heavily reliant on the clinician's expertise, and struggle to objectively quantify the kinematic characteristics of dynamic movement.

Hemiplegic gait represents the most typical manifestation of gait dysfunction. During walking, patients exhibit external rotation of the hip in the swing phase to compensate for restricted knee flexion, resulting in the characteristic "circling gait". In the stance phase, weak ankle plantar flexion causes toe dragging, accompanied by a reduced stride length, decreased step frequency, and symmetric reduction in stride length on both sides. These abnormalities significantly increase fall risk because of the combined effects of impaired balance, reduced stance time, bilateral limb strength and coordination deficits, and decreased walking speed. Thus, vision-based gait analysis technology holds significant value for tracking disease progression and assessing severity [82]. In gait analysis, skeletal landmark sequence data accurately describe human posture and provide key joint position information, forming a crucial foundation for gait analysis and evaluation [83]. Furthermore, optical flow techniques are gaining increasing attention in gait analysis because they enable gait analysis by calculating the relative motion between adjacent frames without requiring prior background information [84]. They capture subtle differences in gait patterns and provide supplementary information for monitoring and evaluating stroke patients' gait. With the rapid advancement of multimodal fusion techniques, the limitations of single-modality data have become increasingly apparent. Multimodal data fusion, which integrates features extracted from various modalities, significantly enhances the recognition performance of gait dysfunction [32,85].

3.1. Skeletal keypoint-based features

Pose estimation is a computer vision-based technique that predicts and localizes key human skeletal landmarks from single images or image sequences (Fig. 1). This technique outputs either static skeletal models or continuous sequences of skeletal keypoints, providing data support for dynamic gait quantification analyses. The ongoing advancement of skeleton-point extraction algorithms, including OpenPose [86], AlphaPose [87], and DeepLabCut [88], has significantly improved the accuracy, efficiency, and dimensionality of the resulting sequence data. This progress provides a solid foundation for constructing a more comprehensive and multidimensional gait quantification parameter system.

OpenPose employs a bottom-up estimation paradigm, achieving real-time whole-body human pose estimation by simultaneously predicting joint confidence maps and part affinity fields (PAFs). This technology captures human motion information from videos and provides an effective tool for quantifying clinical gait analysis. Post-stroke patients often experience secondary changes such as loss of coordination and restricted joint mobility due to primary factors such as decreased muscle strength and neuromuscular control disorders [89, 90], ultimately manifesting as characteristic gait dysfunctions such as hemiplegic gait. Zhou et al. [82] investigated the severity grading of hemiplegic gait based on skeletal keypoint sequence data extracted by OpenPose. They constructed a lightweight attention-based CNN

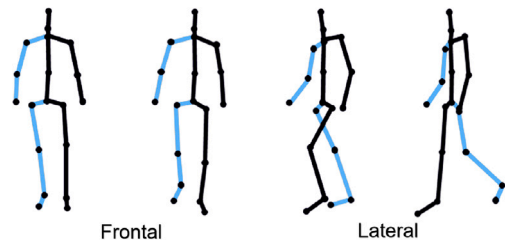

Fig. 1. Frontal and lateral views of the human skeletal keypoints. The two poses on the left depict the skeletal structure of the human body from the front, reflecting its symmetry. In contrast, the two poses on the right show the skeletal structure from the side, highlighting the bending of the legs and the forward and backward tilt of the body.

model for severity classification. The classification model evaluates factors such as circumduction, limited knee flexion during hip extension, shortened double-support phase in the gait cycle, and slowed movement velocity of the affected upper and lower limbs. However, the study primarily focused on evaluating system construction and classification performance enhancements without detailing how these specific features are quantitatively extracted and computed from raw skeletal keypoint time series data. Stroke-induced hemiplegia often leads to impaired balance and motor asymmetry [91], manifesting not only as abnormal gait parameters such as a shortened support phase and bilateral asymmetry but also as reduced walking speed, significantly increasing patients' fall risk [21]. For fall risk assessment, Chen et al. [92] employed skeletal keypoint coordinates extracted via OpenPose and applied Euler's theorem to calculate limb joint angles. By analyzing patients' movement performance and completion times while sitting, sit-to-stand transitions, and standing, they comprehensively evaluated lower-limb muscle strength, balance control capabilities, and fall risk levels. To more precisely capture temporal dynamics in gait, NG et al. [93] derived multiple temporal gait parameters (e.g., step frequency, gait symmetry index, step-time coefficient of variation) and spatial relative distance metrics (e.g., average step width, step-width coefficient of variation, stability margin) from OpenPose extracted 2D skeletal keypoint sequences. This enabled automated gait assessment for fall risk prediction. The study primarily examined the associations between these quantitative features extracted from skeletal keypoint sequences and clinical functional ability scores, as well as future fall incidence rates. By contrast, the method proposed by NG et al. offers a more comprehensive and dynamic assessment by extracting continuous temporal and spatial parameters. Changes in the knee joint angle serve as crucial indicators for evaluating hemiplegia severity [94]. On the basis of this theory, Nishizawa et al. [95] compared 2D keypoints extracted via OpenPose with 3D keypoints from an OptiTrack marker-based system for gait assessment. Their results demonstrated that knee angle data derived from OpenPose exhibited high validity and reliability. The method effectively captured key kinematic parameters of the knee joint, including the maximum extension angle, maximum flexion angle, and range of motion, thereby establishing a reliable basis for distinguishing healthy individuals from hemiplegic patients. Hip and knee muscle weakness in stroke patients has been shown to reduce walking speed [96], a finding further corroborated by Yamamoto et al. in a comparative study [97]. Although the OpenPose-based system performs well in measuring hip and knee kinematics in the sagittal plane and is suitable for analysis under various gait conditions, its accuracy in ankle-angle measurement remains comparatively limited. With respect to ankle dysfunction, bidirectional stiffness training is effective in improving patients' static and dynamic balance control capabilities and gait speed [98]. Building upon the validation of joint angle measurements, Yamamoto et al. [99] further employed OpenPose-derived skeletal keypoint sequences to compute spatiotemporal gait parameters such as walking speed, step frequency, stride length, and

Table 3
Clinical applications of vision-based gait analysis using pose estimation in stroke care.

Study	Algorithm	Clinical application	Extracted gait features
Zhou et al. [82]	OpenPose	Classification of hemiplegic gait severity levels based on spasticity degree	• Circumduction • Limited knee flexion during hip extension • Reduced duration of the double-support phase • Slowed movement velocity on the affected side
Chen et al. [92]	OpenPose	Evaluation of lower limb muscle strength, balance control capacity, and stratifies fall risk levels	• Joint angles • Completion time (sitting, sit-to-stand transitions, standing) • Motor performance
NG et al. [93]	OpenPose	Automated assessment and fall prediction	• Step length/step time variability coefficient • Gait Symmetry Index • Mean step width • Step width variability coefficient • Stability margin
Nishizawa et al. [95]	OpenPose	OpenPose data valuable for clinical diagnosis and evaluation	• Maximum extension angle • Maximum flexion angle • Range of motion
Yamamoto et al. [99]	OpenPose	Validate the effectiveness of hip and knee joint measurements, Analyze the impact of AFO stiffness on gait parameters and optimize rehabilitation programs	• Walking speed • Step frequency • Stride length • Durations of the support and swing phases
Bejinariu et al. [100]	OpenPose	Rehabilitation level assessment	• Thigh midline alignment • Angle between the medial thigh and the horizontal plane • Angle between the medial lower leg and the horizontal plane • Angle between the hip and the medial side of the thigh
Goyal et al. [103]	AlphaPose	Abnormal gait detection and disease diagnosis	• Limb straightness • Hand–leg coordination • Upper-body straightness • Central distances • Mutual distances
Moro et al. [26]	DeepLabCut	Estimation of stroke clinical parameters	• Step length • Stride time • Stance phase • Swing phase
Lonini et al. [28]	DeepLabCut	Estimation of stroke clinical parameters	• Step frequency • Double support time • Swing time • Stance time • Walking speed

durations of the support and swing phases. Through analysis of the effects of ankle foot orthoses (AFOs) with varying stiffness levels on these parameters, they demonstrated that optimizing AFO stiffness can effectively enhance walking function. This provides a quantitative basis for personalized gait rehabilitation via accessible visual technology for patients with neurological conditions such as stroke. In contrast, Bejinariu et al. [100] focused on static and quasistatic analysis of leg posture in patients, and utilized OpenPose-derived skeletal keypoint data extracted from frontal-plane walking videos to compute morphological indicators for assessing rehabilitation progress. These metrics included the femoral midline–knee ankle alignment, the femoral medial angle relative to the horizontal angle, the tibial medial angle relative to the horizontal angle, and the hip-to-femoral medial angle. However, purely morphological analysis struggles to reveal functional-level impairments. Research indicates that asymmetry in swing and stance phase durations during gait is closely associated with abnormal activation of ankle muscle groups [101], whereas reduced weight-bearing on the paralyzed side during static standing directly reflects impaired support function [102]. Therefore, integrating the morphological assessment framework established by Bejinariu et al. with dynamic spatiotemporal parameters and electromyographic signals, which capture muscle function and motor control, holds strong potential for advancing the development of a more precise and reliable rehabilitation assessment model.

In contrast to bottom-up methods, AlphaPose employs a top-down processing pipeline (first detecting human bounding boxes and then estimating keypoints within them), which enhances robustness to variations in background/foreground and lighting conditions and significantly improves joint detection performance. By building on the extracted 2D skeleton point sequences, researchers have constructed more complex analysis models. For example, Goyal et al. [103] utilized the 2D joint coordinate output by AlphaPose to compute six key gait spatial features: limb straightness, hand leg coordination, upper-body straightness, body straightness, central distances, and mutual distances. By calculating the mean and variance of these features over time, they constructed video-level feature vectors that characterize overall gait patterns. These features were used to train machine learning classification models capable not only of detecting gait abnormalities but also of identifying neurological disorders such as Parkinson's disease [103]. DeepLabCut (DLC), initially widely applied in animal behavior research, is an advanced motion-tracking algorithm derived from DeeperCut [104], that enables target-specific feature extraction by fine-tuning residual networks. It was initially widely applied in animal behavior research. Moro et al. [26] successfully applied a markerless approach to human gait analysis in stroke patients, estimating spatiotemporal parameters such as step length, stride time, and stance and swing phase durations. These results were validated against a marker-based motion capture system. The study revealed no statistically significant differences between the systems for most parameters, with the exception of relatively larger errors in the ankle and foot regions. Similarly, Lonini et al. [28] fine-tuned the neural network of the DeepLabCut framework to track skeletal keypoints, including the hip, knee, ankle, heel, and toe, in patient videos. They subsequently trained a CNN model to estimate clinical gait parameters from the trajectories of these skeletal keypoints.

The above research focused primarily on applying two-dimensional pose estimation techniques to gait analysis in stroke patients, with the relevant clinical assessment parameters summarized in Table 3. Each study aims to achieve a quantitative assessment of post-stroke gait disorders through computer vision methods and to apply these insights to clinical diagnosis, rehabilitation monitoring, and fall risk prediction. The extracted features cover multiple dimensions, including joint angles, spatiotemporal parameters (step length, cadence, and stance/swing phase duration), symmetry indices, and dynamic stability parameters. For a more accurate assessment of motor function, the field is progressively migrating to 3D pose estimation. Such approaches typically employ deep neural networks to reconstruct more spatially informative 3D skeletal motion models from 2D joint coordinate data. Among the relevant model architectures, RNNs and graph convolutional networks (GCNs) [105] are the most widely applied. RNNs effectively model the temporal dynamics of skeletal joint data, making them suitable for extracting time-dependent patterns in motion [106, 107]. Graph neural networks (GNNs), inherently suited for graph-structured data, explicitly model the connectivity between joints and bones while enabling hierarchical feature learning through skeletal topology [108]. In rehabilitation assessment research [83,109], GNNs enhance evaluation performance by explicitly capturing spatial relationships between skeletal joints during rehabilitation training. This enables the construction of gait graph models that better align with motor physiology (e.g., by introducing gait-specific links to strengthen joint associations [83]), thereby improving assessments of movement quality and functional recovery in rehabilitation patients. However, in clinical and daily health-monitoring scenarios, gait graph models built with GNNs still face challenges of insufficient sensitivity and specificity. This limitation may lead to missed detections of critical movement abnormalities or false positives in normal movements, thereby compromising the reliability of assessment results.

3.2. Optical flow-based features

Optical flow is a crucial method in motion estimation, analyzing changes in pixel brightness between consecutive image frames to estimate the direction and velocity of an object within a scene, as illustrated in Fig. 2. Because optical flow reveals the instantaneous velocity of pixel motion in images and typically requires no prior scene knowledge, it can accurately capture information such as changes in human body position. Although traditional optical flow computation frameworks [110,111] have achieved significant progress, they often face bottlenecks in practical applications because of high computational complexity and performance limitations [112]. By capturing the dynamic changes in body parts during walking, optical flow provides valuable information for motion pattern analysis and serves as an essential complement to the spatiotemporal features derived from static images.

Kobayashi et al. [113] proposed a method for processing gait videos and extracting gait features while protecting pedestrian privacy. The approach first performs edge detection and optical flow computations on the foot region within the video and then clusters pixels based on distance to identify candidate foot regions. By analyzing acceleration optical flow information within these candidate regions, frames corresponding to heel strike events can be identified. Simultaneously, edge detection and optical flow features are combined to search for static regions, enabling the detection of toe-off moments. This method relies solely on foot region characteristics, reducing the risk of exposing patients' personal information. Moreover, it demonstrates consistent performance in detecting heel-strike and toe-off events across both cluttered and clean backgrounds, with no significant differences, highlighting its strong environmental adaptability and clinical applicability. On the basis of the decomposition of gait optical flow images (GOFI), Luo et al. [84] utilized vertical motion variations in the optical flow field to detect complete gait cycles. Hawas et al. [29] extracted

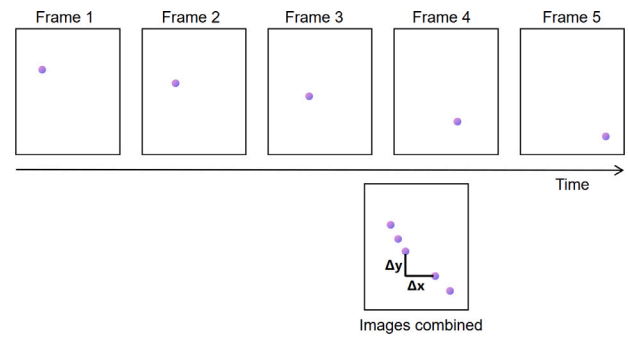

Fig. 2. Motion trajectory and optical flow information of a feature across successive frames. There are five frames, each depicting the feature's location. As time progresses, the feature's position changes, creating a motion trajectory. The lower section of the figure displays the displacement of the feature across multiple images, indicating the changes along the x and y axes ($\triangle x$ and $\triangle y$). This information aids in analyzing the direction and speed of the motion of the optical flow feature.

features from contour image sequences to construct gait energy images (GEIs) [114], analyzing their optical flow to capture dynamic characteristics. The incorporation of optical flow into the GEI framework enhances the analysis and interpretation of gait dynamics. Ye et al. [30] proposed a gait optical-flow imaging method that integrates instantaneous motion direction and intensity information into the original gait contour, reducing reliance on contours as the sole predictive input and thereby improving gait-analysis accuracy. Mahfouf et al. [115] extracted motion features for gait recognition based on optical flow estimated from consecutive frames and computed local optical flow features, global optical flow features, and optical flow histogram features. Among these, local optical flow features capture fine details and local deformations, enhancing the ability to distinguish local actions; global optical flow features describe overall motion patterns and postural changes, complementing the limitations of local features; and optical flow histogram features provide a robust statistical description, facilitating subsequent classification analysis. Although these four studies [29,30,84,115] are not directly targeted at stroke diagnosis and rehabilitation, the optical flow–based dynamic feature extraction methods they employ demonstrate broad applicability in medical and clinical research. Such methods offer objective, quantitative tools for assessing gait stability and periodic disturbances in stroke patients, thereby aiding in the precise identification of specific motor dysfunction patterns.

During normal walking, optical flow information aligns with proprioceptive feedback [116]. Studies [117,118] indicate that when the optical flow velocity mismatches proprioceptive feedback from the legs, individuals adjust their walking speed to reduce this discrepancy. The influence of optical flow velocity on movement has been validated in healthy populations (particularly elderly individuals) and in certain clinical groups [116]. Lamontagne et al. [119] investigated the relationship between reduced walking speed after stroke and perceived optical flow velocity. They compared how stroke patients and healthy individuals adjusted their walking speed in response to changes in optical flow velocity and explored whether a virtual environment manipulating optical flow velocity could facilitate voluntary changes in walking speed among stroke patients. It is generally accepted that stroke patients can still utilize optical flow information while walking, although this ability may be altered and responses may vary among individuals depending on the location of brain damage [120]. Understanding these optical flow patterns is crucial for tailoring rehabilitation strategies to meet each patient's unique needs. Throughout this process, the human body's center of mass (COM) must remain within an appropriate range to maintain balance. If swaying causes the COM to deviate from its equilibrium position, it can be restored

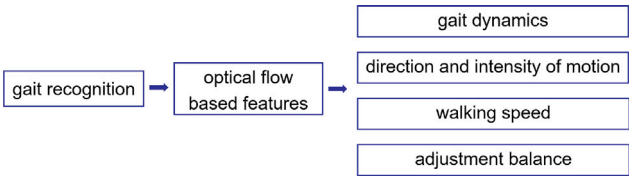

Fig. 3. Through the analysis of optical flow characteristics, individual gait features can be understood and recognized more comprehensively. Gait dynamics reflect the overall movement pattern of an individual during walking; the direction and intensity of movement help analyze the details of gait; walking speed is an important index in gait analysis, effectively quantifying an individual's moving speed; and balance adjustment assesses an individual's ability to maintain balance while walking, reflecting the stability of their movement.

to an appropriate location by adjusting the step length, accelerating or decelerating step speed, or changing movement direction [121]. Step-length adjustments can be reflected in abnormal changes in the optical flow field [121]. By detecting such abnormal changes in the optical flow field, Song et al. [122] investigated balance impairments in patients with chronic ankle instability (CAI) and identified stride width, stride length, and sacral medial/lateral movement as key factors affecting walking balance in patients with CAI. Abnormal changes in the optical flow field provide dynamic visual information that aids patients with CAI in spatial orientation and gait rate adjustment during walking. Furthermore, hemiplegia following stroke can alter lower limb biomechanics, increasing fall risk [123]. Consequently, the ankle joint is crucial for postural stability and locomotor balance control. In this context, optical flow can play a supportive role by providing real-time visual feedback, assisting stroke patients in better perceiving their surroundings and thereby enhancing spatial orientation and gait stability.

Although optical flow can elucidate motion principles through pixel displacement between frames, changes in lighting conditions within the optical-flow field significantly affect the brightness of displaced pixels [124]. Such variations do not necessarily represent actual motion, potentially leading to misalignments and misinterpretations of movement [124]. Furthermore, optical-flow techniques must demonstrate robustness against noise and other imaging artifacts commonly found in real world data [124]. Despite its potential in motion estimation, current research on its impact on stroke survivors' movement remains limited [119,125]. Given the pivotal role of optical-flow estimation in understanding and analyzing abnormal gait patterns, Fig. 3 illustrates optical flow characteristics obtained through feature analysis alongside multiple gait related dynamic features. Further clinical research and validation are required to confirm the efficacy of these features in promoting recovery and rehabilitation among stroke patients. Moreover, actively exploring how variations in optical flow velocity influence gait patterns in post-stroke individuals is crucial. Such insights hold promise for advancing the development of more personalized and efficient rehabilitation strategies.

3.3. Fusion-based features

Most current automated gait analysis systems struggle to perform qualitative and quantitative assessments of gait abnormalities simultaneously [126], often relying on a single data source. This limitation hinders their ability to provide comprehensive and accurate functional evaluations in clinical practice. With the significant expansion of the data scale and the increasing diversity of data types, research is shifting toward heterogeneous multimodal data fusion. This approach aims to enhance model generalizability and robustness in complex real world clinical scenarios by integrating diverse information streams.

In human gait recognition research, contour- and skeleton-based approaches are the two most commonly used methods. Contour information is susceptible to interference from factors such as clothing and occlusion, whereas skeleton data often lack appearance features such as individual body shape. To overcome the limitations of single-modality approaches, current research focuses predominantly on fusing skeleton and contour features [31,32,127] to construct more comprehensive and robust gait representations. Cui et al. [31] proposed a transformer-based gait recognition framework that effectively integrates spatiotemporal features from both contour and skeleton data, addressing the shortcomings of relying on a single data source. Skeletal data capture joint trajectories and angles, whereas contours reflect trunk displacement and limb swing; their deep integration enhances the robustness of gait analysis and assessment, holding potential value for multidimensional feature evaluation in stroke patients. Zeng et al. [32] developed a neural network model based on skeletal-contour information fusion to achieve automated quantitative assessment of gait disorders in patients with Parkinson's disease. Archila et al. [128] proposed a method that integrates gait and eye fixation information. By combining kinematic features extracted from dense optical flow with depth features extracted via a CNN during the early and late fusion stages, they detected of Parkinson's related movement abnormalities. These studies not only provide effective tools for assessing motor function in patients with Parkinson's disease but also demonstrate the potential and clinical applicability of multimodal fusion in stroke rehabilitation evaluation, laying the foundation for extending this approach to gait analysis across broader neurological disorders. The GEI integrates multiple frames of human contours across a complete gait cycle to generate an average energy representation that preserves the overall shape and movement patterns, effectively enhancing the static structural information of body shape. Similarly, the skeletal energy image aggregates joint trajectory data to construct a structured representation highlighting kinematic characteristics. By fusing these two energy images, Loureiro and Correia [129] achieved effective differentiation of pathological gait types, demonstrating the complementary value of multimodal static representations in gait analysis. Beyond skeletal keypoint sequences, optical flow captures minute motion information between frames and exhibits strong robustness to variations in viewpoint and clothing. Feng et al. [130] proposed a novel multimodal gait recognition framework, GaitFusion, which integrates contour features with optical flow features. To uncover latent commonalities and related attributes between the two modalities, they designed a transformer-based bottleneck fusion feature extractor. Additionally, a motion feature enhancer was introduced. By amplifying motion features and incorporating a channel attention mechanism to suppress redundant information, it effectively enhances feature discriminative power and multi-modal fusion performance. In contrast, Cao et al. [131] proposed a fall detection method based on the fusion of skeleton images and dense optical flow sequences. Both approaches are applicable to stroke gait recognition tasks and hold significant practical value for preventing falls caused by balance dysfunction in stroke patients.

4. Discussion

4.1. Technical and clinical limitations

- **Light sensitivity and insufficient algorithm generalization capability in facial analysis**
  Vision-based facial analysis methods are highly sensitive to variations in ambient lighting. Strong light or shadows on one side of the face introduce significant illumination gradients in the image, and this interference is difficult to eliminate completely during both 2D image processing and 3D reconstruction [132]. Because robust algorithms capable of effectively distinguishing grayscale differences caused by external lighting changes from

physiological signals that truly originate from muscle activity or skin texture are still lacking, models often misinterpret lighting artifacts as pathological features such as facial muscle asymmetry, leading to distorted assessment results.

- **Disconnect between kinematic parameters and neuromuscular-physiological indicators in gait analysis**
  Vision-based gait analysis can effectively capture kinematic parameters such as step frequency, stride length, and joint range of motion [133]. However, it cannot directly measure key mechanical and neurophysiological indicators, including muscle activation intensity, tension distribution, ground reaction forces, and spasticity levels. This limitation hampers a comprehensive understanding of post-stroke neuromuscular control mechanisms and limits their use in precision rehabilitation assessment. To achieve a thorough assessment of motor function, future research should explore multimodal fusion analysis frameworks that integrate visual sensing with wearable technologies, such as surface electromyography and IMU.
- **Multimodal data scarcity and interdisciplinary integration challenges**
  Insufficient data represent a core bottlenecks limiting development in this field. Current data resources for stroke research predominantly focus on single modalities, such as standalone facial expression databases [70,134] or abnormal gait datasets [135], while lacking publicly available multimodal datasets containing synchronized facial and gait information. Owing to patient privacy and data security restrictions, most high-quality proprietary datasets are difficult to share [136], resulting in inadequate validation of competition algorithms regarding authenticity, generalization, and clinical applicability. Furthermore, collecting multimodal data requires stricter ethical review, whereas the technical aspects present significant challenges in the spatiotemporal alignment and fusion of multisource heterogeneous data. From a research paradigm perspective, facial neuromuscular control disorders and gait abnormalities in stroke patients are often studied separately within distinct disciplines such as neurology and rehabilitation engineering. This disciplinary barrier contributes to fragmented and decentralized data collection efforts, hindering the effective fusion and comprehensive utilization of resources.

4.2. Bridging vision-based technologies and clinical care

Vision-based technology must emphasize clinical operability, with its core focus on seamlessly integrating technical outputs into clinical workflows and providing clear guidance for diagnostic decisions and patient interventions. Although such technologies can generate highly accurate, repeatable quantitative metrics, such as joint range of motion, trajectory analysis, movement velocity, and symmetry indices, their practical value remains limited if these data cannot be translated into actionable clinical recommendations (e.g., whether referral to a specialist is warranted, or if rehabilitation training intensity or frequency should be adjusted). In this sense, traditional machine learning and deep learning approaches should be viewed as digital extensions and enhancements of traditional scoring systems such as the Fugl-Meyer assessment [81] and the HB grading system [36]. They not only provide finer, continuous objective measurements but also must be embedded within clinical pathways. This integration forms a closed loop system with diagnosis, treatment planning, and efficacy assessment, thereby advancing personalized rehabilitation management models.

However, the safety of computer vision–based auxiliary diagnosis technology remains a core concern for both clinicians and patients. These concerns primarily stem from two aspects: first, insufficient model interpretability, where deep learning decision-making processes are often regarded as “black boxes” [137], making it difficult for clinicians to understand their reasoning; and second, external validity, specifically whether models trained on particular datasets can maintain robust performance across different populations, healthcare institutions, and even different devices. To enhance clinical trust and acceptance, interpretability frameworks such as SHAP (SHapley Additive exPlanations) can be introduced to provide rational explanations for model predictions, aiding clinicians in understanding the basis for judgments [138]. Additionally, establishing mechanisms for continuous model calibration and performance monitoring is essential to ensure the long-term stability and reliability of the system.

The impact of this technology on clinicians’ workload in real clinical settings requires urgent evaluation. Its deployment should aim to reduce repetitive tasks for healthcare professionals rather than increase operational complexity or interpretation burdens. Simultaneously, the system faces significant ethical and compliance challenges. For example, implementing continuous video monitoring in hospital settings involves the collection and processing of sensitive biometric information. This necessitates strict adherence to applicable data protection and privacy regulations within the relevant country or region. It is essential to implement comprehensive security mechanisms such as data anonymization, encrypted storage, and tiered access controls while also ensuring that full informed consent is obtained from patients.

Moreover, owing to the highly sensitive nature and privacy requirements of medical data, the phenomenon of “data silos” is prevalent [139], particularly the lack of high quality, multicenter, multimodal datasets for stroke patients that include synchronized facial and gait information. This severely limits the training effectiveness and generalizability capabilities of models. Finally, driving the practical implementation of these technologies requires robust talent support: cultivating a cohort of interdisciplinary professionals who master artificial intelligence techniques while deeply understanding clinical needs is essential to bridging technological development with medical applications.

4.3. Vision-based alternatives to wearable sensor challenges

In the clinical application of gait analysis, research is gradually shifting its focus from traditional wearable sensor technology to vision-based methods. Owing to its noncontact and minimally invasive characteristics, the latter has emerged as a highly promising alternative. Although wearable sensor technology can precisely measure gait dynamics and kinematics [129], its application faces several limitations: such devices are intrusive, causing patient discomfort and potentially interfering with natural gait owing to their size, weight, or attachment location, which can introduce measurement bias [129]; laboratory constraints and surface mounted devices may mask subtle yet critical variations in gait patterns [140]. Furthermore, patients with certain neurological conditions may unconsciously adjust their gait or conceal movement disorders when aware of monitoring, further increasing the risk of data distortion [141]. Vision-based gait analysis offers an effective solution to these challenges. This approach requires only standard cameras (e.g., RGB cameras) or even smartphones to capture gait videos in natural settings. Computer vision algorithms (such as OpenPose-based human skeletal keypoint detection) enable noncontact, unobtrusive extraction of gait parameters, including spatiotemporal and kinematic metrics such as step speed, stride length, step frequency, joint ranges of motion, the stance-swing phase ratio, and trunk symmetry. Moreover, optical flow estimation techniques can capture overall motion patterns and dynamic stability, enabling fine-grained analysis of gait sequences. Crucially, visual methods support multimodal fusion architectures that integrate multiview videos and depth images, enhancing model robustness and interpretability through feature-level or decision-level fusion. Vision-based systems offer core advantages in the term of implementation convenience, low cost, and strong scalability, as they do not rely on specific experimental environments or require cumbersome device wear and calibration [19,69]. This approach is particularly suitable for rapid screening in clinical settings, home-based remote monitoring, and mid- to long-term rehabilitation tracking [140]. This approach provides a more natural and ecologically valid technical pathway for gait function assessment and personalized rehabilitation interventions in stroke patients.

4.4. Future directions

Facial and gait analysis originated from two distinct branches of computer vision research, each with its own traditional applications and technological development paths. Early facial analysis technology was primarily applied to human computer interaction, security, and biometric identification, with a focus on the extraction of static or near-static biometric features and identity verification. By contrast, gait analysis has historically focused on human motion modeling, behavior recognition, and health status monitoring, emphasizing the analysis of movement patterns within dynamic sequences. Owing to substantial differences in research objectives, data characteristics, evaluation metrics, and acquisition equipment, previous attempts to integrate these two fields into a unified research framework have been limited. However, central nervous system damage caused by stroke often simultaneously triggers facial motor dysfunction, such as facial paralysis and facial asymmetry, and gait abnormalities, such as hemiplegic gait and balance disorders. This category of disorders presents a significant opportunity for the convergence of these two technologies. Facial and gait analyses exhibit strong technical complementarity: facial features reflect the loss of fine motor control, whereas gait data reveal deterioration in overall movement coordination and stability. These modalities can be fused at the feature level to construct multimodal biomarkers or integrated at the decision layer to enhance overall assessment performance. Punyani et al. [142] exemplify this with their dual-layer feature fusion approach, which significantly improves age estimation accuracy by integrating facial and gait features. Although they focus on age inference, their fusion framework provides valuable insights for the collaborative analysis of multimodal medical images and behavioral signals. A similar approach could be adapted for stroke assisted diagnosis and rehabilitation assessment tasks. By integrating static and dynamic features, such as facial asymmetry and eye–mouth coordination, with spatiotemporal parameters and symmetry indices from gait cycles, more comprehensive, sensitive, and reliable assessment models can be constructed. These models play a vital role in disease staging, rehabilitation progress tracking, and evaluation of treatment efficacy.

Notably, both facial and gait data can be conveniently collected via noncontact devices such as standard cameras or smartphones, which offers significant advantages in terms of accessibility, efficiency, and scalability. In community screening and primary health care settings, medical staff can conduct preliminary screenings for facial and gait abnormalities via standard camera equipment without relying on expensive, specialized devices. This significantly lowers the threshold and cost of large-scale stroke risk screening, supporting the public health goal of "early detection and early intervention". Furthermore, for stroke patients requiring prolonged rehabilitation and regular follow-up, the traditional hospital centered rehabilitation model faces implementation challenges due to the need for frequent visits. In medically underserved remote areas, including mountainous regions and island communities, many patients struggle to adhere to long-term standardized rehabilitation because of transportation difficulties and limited access to rehabilitation resources, which ultimately compromises functional recovery outcomes. Vision-based facial and gait analysis technology offers a viable solution to these challenges. By continuously capturing subtle recovery changes in facial movements and improvement trends in gait parameters, the system enables multimodal fusion analysis. It can automatically generate rehabilitation progress reports and issue timely risk alerts, thereby enhancing the continuity, accessibility, and adherence of rehabilitation management while reducing reliance on specialized equipment and in-person medical resources. Consequently, tele-rehabilitation scenarios that integrate facial and gait analysis technologies are emerging as a significant direction for chronic stroke management and rehabilitation support.

5. Conclusion

This review systematically examined research advances in computer vision-based technologies for facial and gait analysis in stroke diagnosis, treatment, and rehabilitation. We explored applications ranging from traditional machine learning to deep learning architectures for identifying facial asymmetry and motor unit characteristics. Additionally, we detailed feature extraction methods, including skeletal keypoints and optical flow, for quantifying spatiotemporal and kinematic parameters of gait. Analysis indicates that these noninvasive, low cost vision-based technologies offer viable alternatives to traditional approaches reliant on subjective assessments and wearable sensor-based technologies, particularly for primary care screening, long-term home rehabilitation monitoring, and tele-rehabilitation scenarios. However, their clinical translation faces multifaceted challenges: technologically, issues such as algorithm generalization, sensitivity to lighting conditions, and black box decision mechanisms remain unresolved. In terms of data, the scarcity of high-quality, multimodal clinical datasets severely hinders model advancement. Clinically, seamlessly integrating these outputs into existing diagnostic workflows and translating them into actionable clinical decisions will ultimately determine their practical value.

Despite existing challenges, vision-based stroke diagnosis, treatment, and rehabilitation research holds great promise. Future research will focus on building multimodal fusion frameworks that integrate diverse information sources such as visual data, speech, and electronic medical record texts; developing more efficient, lightweight models and edge computing solutions to support real-time remote monitoring and rehabilitation guidance in community and home settings; and advancing explainable artificial intelligence (XAI) research to provide not only predictive outputs but also transparent, trustworthy decision support for clinicians. We firmly believe that through deep interdisciplinary collaboration and integration, computer vision-based technology will become an indispensable component of stroke care. It will continuously bridge technological innovation with clinical needs, providing core momentum for achieving inclusive, precise, and efficient stroke management.

CRediT authorship contribution statement

Xiao Han: Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Li-Ping Li:** Writing – original draft, Data curation, Conceptualization. **Zi-Yan Wang:** Writing – original draft, Visualization, Data curation. **An-Ying Xu:** Writing – original draft, Investigation, Data curation. **Tian-Shu Wang:** Writing – review & editing, Formal analysis. **Tao Yang:** Writing – review & editing, Formal analysis. **Kong-Fa Hu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgments

We would like to thank the instructors and students of the project team for their support, especially Prof. Yawei Song from Nanjing Sports Institute of Exercise Science and Health Promotion for his guidance in wearable gait rehabilitation, and our family members for their full support.

Data availability

No data was used for the research described in the article.

References

[1] Stroke Overview. https://www.ninds.nih.gov/health-information/stroke/stroke-overview.

[2] Stephen J.X. Murphy, David J. Werring, Stroke: causes and clinical features, Medicine 48 (9) (2020) 561–566.

[3] A.S. Update, Heart disease and stroke statistics–2017 update, Circulation 135 (e146-e603) (2017) 1.

[4] World Stroke Day 2022. https://www.who.int/srilanka/news/detail/29-10-2022-world-stroke-day-2022.

[5] Karl-Olof Lövblad, Stephen Altrichter, Vitor Mendes Pereira, Maria Vargas, Ana Marcos Gonzalez, Sven Haller, Roman Sztajzel, Imaging of acute stroke: CT and/or MRI, J. Neuroradiol. 42 (1) (2015) 55–64.

[6] Pedro Vilela, Howard A. Rowley, Brain ischemia: CT and MRI techniques in acute ischemic stroke, Eur. J. Radiol. 96 (2017) 162–172.

[7] Rashmi U. Kothari, Arthur Pancioli, Tiepu Liu, Thomas Brott, Joseph Broderick, Cincinnati prehospital stroke scale: reproducibility and validity, Ann. Emerg. Med. 33 (4) (1999) 373–378.

[8] Joseph Harbison, Omar Hossain, Damian Jenkinson, John Davis, Stephen J. Louw, Gary A. Ford, Diagnostic accuracy of stroke referrals from primary care, emergency room physicians, and ambulance staff using the face arm speech test, Stroke 34 (1) (2003) 71–76.

[9] Aysen Degerli, Pekka Jäkälä, Juha Pajula, Milla Immonen, Miguel Bordallo López, MAMAF-Net: Motion-aware and multi-attention fusion network for stroke diagnosis, Biomed. Signal Process. Control. 95 (2024) 106381.

[10] Sushanth Aroor, Rajpreet Singh, Larry B. Goldstein, BE-FAST (balance, eyes, face, arm, speech, time) reducing the proportion of strokes missed using the FAST mnemonic, Stroke 48 (2) (2017) 479–481.

[11] Martin Schimmel, T. Ono, O.L.T. Lam, F. Müller, Oro-facial impairment in stroke patients, J. Oral. Rehabil. 44 (4) (2017) 313–326.

[12] Kai Klepzig, Martin Domin, Julia Wendt, Bettina von Sarnowski, Alexander Lischke, Alfons O. Hamm, Martin Lotze, Structural integrity of the insula and emotional facial recognition performance following stroke, Brain Commun. 5 (3) (2023) fcad144.

[13] Jorge Latorre, Carolina Colomer, Mariano Alcañiz, Roberto Llorens, Gait analysis with the Kinect v2: Normative study with healthy individuals and comprehensive study of its sensitivity, validity, and reliability in individuals with stroke, J. Neuroeng. Rehabil. 16 (1) (2019) 97.

[14] Chuan-Yu Chang, Man-Ju Cheng, Matthew Huei-Ming Ma, Application of machine learning for facial stroke detection, in: 2018 IEEE 23rd International Conference on Digital Signal Processing, DSP, IEEE, 2018, pp. 1–5.

[15] Gemma S. Parra-Dominguez, Raul E. Sanchez-Yanez, Carlos H. Garcia-Capulin, Facial paralysis detection on images using key point analysis, Appl. Sci. 11 (5) (2021) 2435.

[16] Jérôme Thevenot, Miguel Bordallo López, Abdenour Hadid, A survey on computer vision for assistive medical diagnosis from faces, IEEE J. Biomed. Heal. Inform. 22 (5) (2017) 1497–1511.

[17] Zolbayar Shagdar, Anette Huuse Farmen, Seyed Ali Amirshahi, Kiran Raja, A survey on computer vision-based automatic assessment of stroke and facial palsy, IEEE Access (2025).

[18] Jiaqi Qiang, Danning Wu, Hanze Du, Huijuan Zhu, Shi Chen, Hui Pan, Review on facial-recognition-based applications in disease diagnosis, Bioengineering 9 (7) (2022) 273.

[19] Andrea Ancillao, Manuela Galli, Eleonora Annese, Simone Criscuolo, Sara L. Vimercati, Domenica Le Pera, Giorgio Albertini, Quantitative evaluation of facial movements in adult patients with hemiplegia after stroke, Int. J. Signal Image Process. 1 (2016).

[20] Sandra M. Woolley, Characteristics of gait in hemiplegia, Top. Stroke Rehabil. 7 (4) (2001) 1–18.

[21] Juan Li, Dongling Zhong, Jing Ye, Mingxing He, Xicen Liu, Hui Zheng, Rongjiang Jin, Shao-lan Zhang, Rehabilitation for balance impairment in patients after stroke: a protocol of a systematic review and network meta-analysis, BMJ Open 9 (7) (2019) e026844.

[22] Judith D. Schaechter, Motor rehabilitation and brain plasticity after hemiparetic stroke, Prog. Neurobiol. 73 (1) (2004) 61–72.

[23] Rafael Caldas, Marion Mundt, Wolfgang Potthast, Fernando Buarque de Lima Neto, Bernd Markert, A systematic review of gait analysis methods based on inertial sensors and adaptive algorithms, Gait Posture 57 (2017) 204–210.

[24] Yoshifumi Kijima, Ryoji Kiyama, Masaki Sekine, Toshiyo Tamura, Toshiro Fujimoto, Tetsuo Maeda, Tadasu Ohshige, Estimation of gait independence using a tri-axial accelerometer in stroke patients, J. Aging Phys. Act. 26 (1) (2018) 61–67.

[25] Mo Wang, Shanshan Yong, Chunjiu He, Hongying Chen, Sixu Zhang, Chen Peng, Xin'an Wang, Research on abnormal gait recognition algorithms for stroke patients based on array pressure sensing system, in: 2019 IEEE 3rd Information Technology, Networking, Electronic and Automation Control Conference, ITNEC, IEEE, 2019, pp. 1560–1563.

[26] Matteo Moro, Giorgia Marchesi, Francesca Odone, Maura Casadio, Markerless gait analysis in stroke survivors based on computer vision and deep learning: A pilot study, in: Proceedings of the 35th Annual ACM Symposium on Applied Computing, 2020, pp. 2097–2104.

[27] Sina Mehdizadeh, Hoda Nabavi, Andrea Sabo, Twinkle Arora, Andrea Iaboni, Babak Taati, Concurrent validity of human pose tracking in video for measuring gait parameters in older adults: a preliminary analysis with multiple trackers, viewing angles, and walking directions, J. Neuroeng. Rehabil. 18 (1) (2021) 139.

[28] Luca Lonini, Yaejin Moon, Kyle Embry, R. James Cotton, Kelly McKenzie, Sophia Jenz, Arun Jayaraman, Video-based pose estimation for gait analysis in stroke survivors during clinical assessments: a proof-of-concept study, Digit. Biomark. 6 (1) (2022) 9–18.

[29] Ahmed R. Hawas, Heba A. El-Khobby, Mohammed Abd-Elnaby, Fathi E. Abd El-Samie, Gait identification by convolutional neural networks and optical flow, Multimedia Tools Appl. 78 (2019) 25873–25888.

[30] Hongyi Ye, Tanfeng Sun, Ke Xu, Gait recognition based on gait optical flow network with inherent feature pyramid, Appl. Sci. 13 (19) (2023) 10975.

[31] Yufeng Cui, Yimei Kang, Multi-modal gait recognition via effective spatial-temporal feature fusion, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 17949–17957.

[32] Qingyi Zeng, Peipei Liu, Ningbo Yu, Jialing Wu, Weiguang Huo, Jianda Han, Video-based quantification of gait impairments in Parkinson's disease using skeleton-silhouette fusion convolution network, IEEE Trans. Neural Syst. Rehabil. Eng. 31 (2023) 2912–2922.

[33] Anping Song, Zuoyu Wu, Xuehai Ding, Qian Hu, Xinyi Di, Neurologist standard classification of facial nerve paralysis with deep neural networks, Futur. Internet 10 (11) (2018) 111.

[34] Yan Zhuang, Mark McDonald, Omar Uribe, Xuwang Yin, Dhyey Parikh, Andrew M. Southerland, Gustavo K. Rohde, Facial weakness analysis and quantification of static images, IEEE J. Biomed. Heal. Inform. 24 (8) (2020) 2260–2267.

[35] Ibrahim Alsharif, Anna Lina Ruscelli, Gabriele Cecchetti, Sujit Kumar Sahu, Molka Gharbaoui, Giovanni Orlandi, Piero Castoldi, Face recognition to support pre-hospital stroke diagnosis, in: 2024 IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events, PerCom Workshops, IEEE, 2024, pp. 227–232.

[36] John W. House, Derald E. Brackmann, Facial nerve grading system, Otolaryngol.—Head Neck Surg. 93 (2) (1985) 146–147.

[37] Martyn J. Fields, Ninian S. Peckitt, Facial nerve function index: a clinical measurement of facial nerve activity in patients with facial nerve palsies, Oral Surg. Oral Med. Oral Pathol. 69 (6) (1990) 681–682.

[38] Facial Nerve Disorders Committee, Jeffrey T. Vrabec, Douglas D. Backous, Hamid R. Djalilian, Paul W. Gidley, John P. Leonetti, Sam J. Marzo, Daniel Morrison, Matthew Ng, Mitchell J. Ramsey, et al., Facial nerve grading system 2.0, Otolaryngol.—Head Neck Surg. 140 (4) (2009) 445–450.

[39] J. Gail Neely, Nevin G. Cherian, Cody B. Dickerson, Julian M. Nedzelski, Sunnybrook facial grading system: reliability and criteria for grading, Laryngoscope 120 (5) (2010) 1038–1045.

[40] Chad M. Aldridge, Mark M. McDonald, Mattia Wruble, Yan Zhuang, Omar Uribe, Timothy L. McMurry, Iris Lin, Haydon Pitchford, Brett J. Schneider, William A. Dalrymple, et al., Human vs. machine learning based detection of facial weakness using video analysis, Front. Neurol. 13 (2022) 878282.

[41] Yan Zhuang, Omar Uribe, Mark McDonald, Xuwang Yin, Dhyey Parikh, Andrew Southerland, Gustavo Rohde, F-DIT-V: An automated video classification tool for facial weakness detection, in: 2019 IEEE EMBS International Conference on Biomedical & Health Informatics, BHI, IEEE, 2019, pp. 1–4.

[42] Pedro F. Felzenszwalb, Ross B. Girshick, David McAllester, Deva Ramanan, Object detection with discriminatively trained part-based models, IEEE Trans. Pattern Anal. Mach. Intell. 32 (9) (2009) 1627–1645.

[43] Wei Wang, Yilin Mo, John A. Ozolek, Gustavo K. Rohde, Penalized fisher discriminant analysis and its application to image-based morphometry, Pattern Recognit. Lett. 32 (15) (2011) 2128–2135.

[44] Josef Finsterer, Management of peripheral facial nerve palsy, Eur. Arch. Otrhinolaryngol. 265 (7) (2008) 743–752.

[45] Karthik Mahadevappa, Ariana Vora, Andrew Graham, Shanker Nesathurai, Facial paralysis: a critical review of accepted explanation, Med. Hypotheses 74 (3) (2010) 508–509.

[46] Amira Gaber, Mona F. Taher, Manal Abdel Wahed, Nevin Mohieldin Shalaby, Sarah Gaber, Classification of facial paralysis based on machine learning techniques, Biomed. Eng. Online 21 (1) (2022) 65.

[47] Amira Gaber, Mona F. Taher, Manal Abdel Wahed, Quantifying facial paralysis using the kinect v2, in: 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC, IEEE, 2015, pp. 2497–2501.

[48] Marti A. Hearst, Susan T. Dumais, Edgar Osuna, John Platt, Bernhard Scholkopf, Support vector machines, IEEE Intell. Syst. Appl. 13 (4) (1998) 18–28.

[49] Zhexiao Guo, Guo Dan, Jianghuai Xiang, Jun Wang, Wanzhang Yang, Huijun Ding, Oliver Deussen, Yongjin Zhou, An unobtrusive computerized assessment framework for unilateral peripheral facial paralysis, IEEE J. Biomed. Heal. Inform. 22 (3) (2017) 835–841.

[50] Hyun Seok Kim, So Young Kim, Young Ho Kim, Kwang Suk Park, A smartphone-based automatic diagnosis system for facial nerve palsy, Sensors 15 (10) (2015) 26756–26768.
[51] Jocelyn Barbosa, Woo-Keun Seo, Jaewoo Kang, paraFaceTest: an ensemble of regression tree-based facial features extraction for efficient facial paralysis classification, BMC Med. Imaging 19 (1) (2019) 30.
[52] Leo Breiman, Random forests, Mach. Learn. 45 (1) (2001) 5–32.
[53] J. Ross Quinlan, Induction of decision trees, Mach. Learn. 1 (1) (1986) 81–106.
[54] Shu He, JohnJ Soraghan, BrianF O'Reilly, Biomedical image sequence analysis with application to automatic quantitative assessment of facial paralysis, EURASIP J. Image Video Process. 2007 (1) (2007) 081282.
[55] Shu He, John J. Soraghan, Brian F. O'Reilly, Dongshan Xing, Quantitative analysis of facial paralysis using local binary patterns in biomedical videos, IEEE Trans. Biomed. Eng. 56 (7) (2009) 1864–1870.
[56] Martin Dietrich Buhmann, Radial basis functions, Acta Numer. 9 (2000) 1–38.
[57] Xulong Liu, Songchao Dong, Menghan An, Luyi Bai, Jingmin Luan, Quantitative assessment of facial paralysis using infrared thermal imaging, in: 2015 8th International Conference on Biomedical Engineering and Informatics, BMEI, IEEE, 2015, pp. 106–110.
[58] Chaoqun Jiang, Jianhuang Wu, Weizheng Zhong, Mingqiang Wei, Jing Tong, Haibo Yu, Ling Wang, Automatic facial paralysis assessment via computational image analysis, J. Heal. Eng. 2020 (1) (2020) 2398542.
[59] Dong Zhang, A method of selecting acupoints for acupuncture treatment of peripheral facial paralysis by thermography, Am. J. Chin. Med. 35 (06) (2007) 967–975.
[60] Jorma Laaksonen, Erkki Oja, Classification with learning k-nearest neighbors, in: Proceedings of International Conference on Neural Networks, ICNN'96, Vol. 3, IEEE, 1996, pp. 1480–1483.
[61] Grace W. Lindsay, Convolutional neural networks as a model of the visual system: Past, present, and future, J. Cogn. Neurosci. 33 (10) (2021) 2017–2031.
[62] Syed Maaher Hossain, Zeeshan Jamal, Aurik Anjum Noshin, Mohammad Monirujjaman Khan, Comparative study of deep learning algorithms for the detection of facial paralysis, in: 2022 IEEE 13th Annual Information Technology, Electronics and Mobile Communication Conference, IEMCON, IEEE, 2022, pp. 0368–0377.
[63] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778.
[64] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, Zbigniew Wojna, Rethinking the inception architecture for computer vision, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2818–2826.
[65] Karen Simonyan, Andrew Zisserman, Very deep convolutional networks for large-scale image recognition, 2014, arXiv preprint arXiv:1409.1556.
[66] Zhexiao Guo, Minmin Shen, Le Duan, Yongjin Zhou, Jianghuai Xiang, Huijun Ding, Shifeng Chen, Oliver Deussen, Guo Dan, Deep assessment process: Objective assessment process for unilateral peripheral facial paralysis via deep convolutional neural network, in: 2017 IEEE 14th International Symposium on Biomedical Imaging, ISBI 2017, IEEE, 2017, pp. 135–138.
[67] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, Andrew Rabinovich, Going deeper with convolutions, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 1–9.
[68] Yi Sun, Xiaogang Wang, Xiaoou Tang, Deep learning face representation from predicting 10,000 classes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 1891–1898.
[69] Gozde Yolcu, Ismail Oztel, Serap Kazan, Cemil Oz, Kannappan Palaniappan, Teresa E. Lever, Filiz Bunyak, Facial expression recognition for monitoring neurological disorders based on convolutional neural network, Multimedia Tools Appl. 78 (22) (2019) 31581–31603.
[70] Cuiting Xu, Chunchuan Yan, Mingzhe Jiang, Fayadh Alenezi, Adi Alhudhaif, Norah Alnaim, Kemal Polat, Wanqing Wu, A novel facial emotion recognition method for stress inference of facial nerve paralysis patients, Expert Syst. Appl. 197 (2022) 116705.
[71] Yiyang Wang, Yunyan Ye, Shengyi Shi, Kehang Mao, Haonan Zheng, Xuguang Chen, Hanting Yan, Yiming Lu, Yong Zhou, Weimin Ye, et al., Prediagnosis recognition of acute ischemic stroke by artificial intelligence from facial images, Aging Cell (2024) e14196.
[72] Pengfei Xu, Fei Xie, Tongsheng Su, Zhaoxin Wan, Zhaoyong Zhou, Xiaoyu Xin, Ziyu Guan, Automatic evaluation of facial nerve paralysis by dual-path LSTM with deep differentiated network, Neurocomputing 388 (2020) 70–77.
[73] Yan Zhuang, Mark M. McDonald, Chad M. Aldridge, Mohamed Abul Hassan, Omar Uribe, Daniel Arteaga, Andrew M. Southerland, Gustavo K. Rohde, Video-based facial weakness analysis, IEEE Trans. Biomed. Eng. 68 (9) (2021) 2698–2705.
[74] François Chollet, Xception: Deep learning with depthwise separable convolutions, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 1251–1258.
[75] Mingxing Tan, Quoc Le, Efficientnet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114.
[76] Rachael L. Cordina, David S. Celermajer, Chronic cyanosis and vascular function: implications for patients with cyanotic congenital heart disease, Cardiol. Young 20 (3) (2010) 242–253.
[77] Ye Ming, Hu Qian, Liu Guangyuan, CNN-LSTM facial expression recognition method fused with two-layer attention mechanism, Comput. Intell. Neurosci. 2022 (1) (2022) 7450637.
[78] Yunfei Lai, A comparison of traditional machine learning and deep learning in image recognition, in: Journal of Physics: Conference Series, Vol. 1314, IOP Publishing, 2019, 012148.
[79] A. Dunn, D.L. Marsden, E. Nugent, P. Van Vliet, N.J. Spratt, J. Attia, R. Callister, Protocol variations and six-minute walk test performance in stroke survivors: a systematic review with meta-analysis, Stroke Res. Treat. 2015 (1) (2015) 484813.
[80] Richard W. Bohannon, A. Williams Andrews, Michael W. Thomas, Walking speed: reference values and correlates for older adults, J. Orthop. Sport. Phys. Ther. 24 (2) (1996) 86–90.
[81] Edgar D. Hernández, Sandra M. Forero, Claudia P. Galeano, Nubia E. Barbosa, Katharina S. Sunnerhagen, Margit Alt Murphy, Intra-and inter-rater reliability of Fugl-Meyer Assessment of Lower Extremity early after stroke, Braz. J. Phys. Ther. 25 (6) (2021) 709–718.
[82] Chengju Zhou, Daqin Feng, Shuyu Chen, Nianming Ban, Jiahui Pan, Portable vision-based gait assessment for post-stroke rehabilitation using an attention-based lightweight CNN, Expert Syst. Appl. 238 (2024) 122074.
[83] Haoyu Tian, Xin Ma, Hanbo Wu, Yibin Li, Skeleton-based abnormal gait recognition with spatio-temporal attention enhanced gait-structural graph convolutional networks, Neurocomputing 473 (2022) 116–126.
[84] Zhengping Luo, Tianqi Yang, Yanjun Liu, Gait optical flow image decomposition for human recognition, in: 2016 IEEE Information Technology, Networking, Electronic and Automation Control Conference, IEEE, 2016, pp. 581–586.
[85] Aite Zhao, Jianbo Li, Junyu Dong, Lin Qi, Qianni Zhang, Ning Li, Xin Wang, Huiyu Zhou, Multimodal gait recognition for neurodegenerative diseases, IEEE Trans. Cybern. 52 (9) (2021) 9439–9453.
[86] Zhe Cao, Tomas Simon, Shih-En Wei, Yaser Sheikh, Realtime multi-person 2d pose estimation using part affinity fields, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 7291–7299.
[87] Hao-Shu Fang, Jiefeng Li, Hongyang Tang, Chao Xu, Haoyi Zhu, Yuliang Xiu, Yong-Lu Li, Cewu Lu, Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time, IEEE Trans. Pattern Anal. Mach. Intell. 45 (6) (2022) 7157–7173.
[88] Alexander Mathis, Pranav Mamidanna, Kevin M. Cury, Taiga Abe, Venkatesh N. Murthy, Mackenzie Weygandt Mathis, Matthias Bethge, DeepLabCut: markerless pose estimation of user-defined body parts with deep learning, Nature Neurosci. 21 (9) (2018) 1281–1289.
[89] Katherine J. Sullivan, David A. Brown, Tara Klassen, Sara Mulroy, Tingting Ge, Stanley P. Azen, Carolee J. Winstein, Physical Therapy Clinical Research Network (PTClinResNet), Effects of task-specific locomotor and strength training in adults who were ambulatory after stroke: results of the STEPS randomized clinical trial, Phys. Ther. 87 (12) (2007) 1580–1602.
[90] Mario Manca, Pia Marchi, A proposal of clinical report regarding dynamic emg in gait disorders following upper motoneuron lesion, Open Rehabil. J. 3 (2010) 151–157.
[91] Sarah F. Tyson, Marie Hanley, Jay Chillala, Andrea Selley, Raymond C. Tallis, Balance disability after stroke, Phys. Ther. 86 (1) (2006) 30–38.
[92] Yen-Chiu Chen, Chia-Jou Yang, Spencer Perkins, Stroke rehabilitation exercise assistance based on deep learning openpose, in: 2023 International Automatic Control Conference, CACS, IEEE, 2023, pp. 1–6.
[93] Kimberley-Dale Ng, Sina Mehdizadeh, Andrea Iaboni, Avril Mansfield, Alastair Flint, Babak Taati, Measuring gait variables using computer vision to assess mobility and fall risk in older adults with dementia, IEEE J. Transl. Eng. Heal. Med. 8 (2020) 1–9.
[94] Karen A. Hughes, Frank Bell, Visual assessment of hemiplegic gait following stroke: pilot study, Arch. Phys. Med. Rehabil. 75 (10) (1994) 1100–1107.
[95] Katsuhiko Nishizawa, Yu Oba, Kento Yamada, Ikumi Tanaka, Toru Tsumugiwa, Ryuichi Yokogawa, Tomoya Watanabe, Evaluation of the clinical utility of a gait analysis system using pose estimation techniques in physical therapy, in: 2024 SICE International Symposium on Control Systems, SICE ISCS, IEEE, 2024, pp. 107–112.
[96] An-Lun Hsu, Pei-Fang Tang, Mei-Hwa Jan, Analysis of impairments influencing gait velocity and asymmetry of hemiplegic patients after mild to moderate stroke, Arch. Phys. Med. Rehabil. 84 (8) (2003) 1185–1193.
[97] Masataka Yamamoto, Koji Shimatani, Masaki Hasegawa, Yuichi Kurita, Yuto Ishige, Hiroshi Takemura, Accuracy of temporo-spatial and lower limb joint kinematics parameters using OpenPose for various gait patterns with orthosis, IEEE Trans. Neural Syst. Rehabil. Eng. 29 (2021) 2666–2675.
[98] Dongyual Yoo, Younsun Son, Dae-Hee Kim, Kap-Ho Seo, Beom-Chan Lee, Technology-assisted ankle rehabilitation improves balance and gait performance in stroke survivors: a randomized controlled study with 1-month follow-up, IEEE Trans. Neural Syst. Rehabil. Eng. 26 (12) (2018) 2315–2323.

[99] Masataka Yamamoto, Koji Shimatani, Daisuke Matsuura, Yusuke Murakami, Naoya Oeda, Hiroshi Takemura, Single camera-based gait analysis using pose estimation for ankle-foot orthosis stiffness adjustment on individuals with stroke, IEEE J. Transl. Eng. Heal. Med. (2025).
[100] Silviu-Ioan Bejinariu, Ramona Luca, Ilie Onu, Gladiola Petroiu, Hariton Costin, Image processing for the rehabilitation assessment of locomotion injuries and post stroke disabilities, in: 2021 International Conference on E-Health and Bioengineering, EHB, IEEE, 2021, pp. 1–4.
[101] Gabriela M. Rozanski, Andrew H. Huntley, Lucas D. Crosby, Alison Schinkel-Ivy, Avril Mansfield, Kara K. Patterson, Lower limb muscle activity underlying temporal gait asymmetry post-stroke, Clin. Neurophysiol. 131 (8) (2020) 1848–1858.
[102] Janna Hendrickson, Kara K. Patterson, Elizabeth L. Inness, William E. McIlroy, Avril Mansfield, Relationship between asymmetry of quiet standing balance control and walking post-stroke, Gait Posture 39 (1) (2014) 177–181.
[103] Daksh Goyal, Koteswar Rao Jerripothula, Ankush Mittal, Detection of gait abnormalities caused by neurological disorders, in: 2020 IEEE 22nd International Workshop on Multimedia Signal Processing, MMSP, IEEE, 2020, pp. 1–6.
[104] Eldar Insafutdinov, Leonid Pishchulin, Bjoern Andres, Mykhaylo Andriluka, Bernt Schiele, Deepercut: A deeper, stronger, and faster multi-person pose estimation model, in: Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, the Netherlands, October 11-14, 2016, Proceedings, Part VI 14, Springer, 2016, pp. 34–50.
[105] Ming Chen, Zhewei Wei, Zengfeng Huang, Bolin Ding, Yaliang Li, Simple and deep graph convolutional networks, in: International Conference on Machine Learning, PMLR, 2020, pp. 1725–1735.
[106] Kooksung Jun, Deok-Won Lee, Kyoobin Lee, Sanghyub Lee, Mun Sang Kim, Feature extraction using an RNN autoencoder for skeleton-based abnormal gait recognition, IEEE Access 8 (2020) 19196–19207.
[107] Thanaporn Jinnovart, Xiongcai Cai, Kundjanasith Thonglek, Abnormal gait recognition in real-time using recurrent neural networks, in: 2020 59th IEEE Conference on Decision and Control, CDC, IEEE, 2020, pp. 972–977.
[108] Lei Shi, Yifan Zhang, Jian Cheng, Hanqing Lu, Skeleton-based action recognition with directed graph neural networks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 7912–7921.
[109] Swakshar Deb, Md Fokhrul Islam, Shafin Rahman, Sejuti Rahman, Graph convolutional networks for assessment of physical rehabilitation exercises, IEEE Trans. Neural Syst. Rehabil. Eng. 30 (2022) 410–419.
[110] Bruce D. Lucas, Takeo Kanade, An iterative image registration technique with an application to stereo vision, in: IJCAI'81: 7th International Joint Conference on Artificial Intelligence, Vol. 2, 1981, pp. 674–679.
[111] Gunnar Farnebäck, Two-frame motion estimation based on polynomial expansion, in: Image Analysis: 13th Scandinavian Conference, SCIA 2003 Halmstad, Sweden, June 29–July 2, 2003 Proceedings 13, Springer, 2003, pp. 363–370.
[112] Syed Tafseer Haider Shah, Xiang Xuezhi, Traditional and modern strategies for optical flow: an investigation, SN Appl. Sci. 3 (3) (2021) 289.
[113] Junya Kobayashi, Nobuaki Nakazawa, Gait event recognition based on edge detection and optical flow from foot video, in: 2024 IEEE International Conference on Robotics and Biomimetics, ROBIO, IEEE, 2024, pp. 45–50.
[114] Jinguang Han, Bir Bhanu, Individual recognition using gait energy image, IEEE Trans. Pattern Anal. Mach. Intell. 28 (2) (2005) 316–322.
[115] Zohra Mahfouf, Hayet Farida Merouani, Imed Bouchrika, Nouzha Harrati, Investigating the use of motion-based features from optical flow for gait recognition, Neurocomputing 283 (2018) 140–149.
[116] Emma De Keersmaecker, Anke Van Bladel, Silvia Zaccardi, Nina Lefeber, Carlos Rodriguez-Guerrero, Eric Kerckhofs, Bart Jansen, Eva Swinnen, Virtual reality—enhanced walking in people post-stroke: effect of optic flow speed and level of immersion on the gait biomechanics, J. NeuroEng. Rehabil. 20 (1) (2023) 124.
[117] Sergei V. Adamovich, Gerard G. Fluet, Eugene Tunik, Alma S. Merians, Sensorimotor training in virtual reality: a review, NeuroRehabilitation 25 (1) (2009) 29–44.
[118] Emma De Keersmaecker, Anke Van Bladel, Silvia Zaccardi, Nina Lefeber, Carlos Rodriguez-Guerrero, Eric Kerckhofs, Bart Jansen, Eva Swinnen, Effects of semi-immersive virtual reality and manipulation of optic flow speed on gait biomechanics in people post-stroke, J. Rehabil. Med. 56 (12384) (2024).
[119] Anouk Lamontagne, Joyce Fung, Bradford J. McFadyen, Jocelyn Faubert, Modulation of walking speed by changing optic flow in persons with stroke, J. Neuroeng. Rehabil. 4 (2007) 1–8.
[120] Emily A. Keshner, Anouk Lamontagne, The untapped potential of virtual reality in rehabilitation of balance and gait in neurological disorders, Front. Virtual Real. 2 (2021) 641650.
[121] Shawn M. O'Connor, Arthur D. Kuo, Direction-dependent control of balance during walking and standing, J. Neurophysiol. 102 (3) (2009) 1411–1419.
[122] Kyeongtak Song, Jason R. Franz, Erik A. Wikstrom, Optical flow balance perturbations alter gait kinematics and variability in chronic ankle instability patients, Gait Posture 92 (2022) 271–276.
[123] Hogene Kim, Jieun Cho, Sangwoo Cho, Joon-Ho Shin, Ankle stiffness asymmetry is associated with balance function in individuals with chronic stroke, Sci. Rep. 13 (1) (2023) 15721.
[124] Andrea Alfarano, Luca Maiano, Lorenzo Papa, Irene Amerini, Estimating optical flow: A comprehensive review of the state of the art, Comput. Vis. Image Underst. 249 (2024) 104160.
[125] Hyung-Kyu Kang, Young Kim, Yijung Chung, Sujin Hwang, Effects of treadmill training with optic flow on balance and gait in individuals following stroke: randomized controlled trials, Clin. Rehabil. 26 (3) (2012) 246–255.
[126] Chengkun Cui, Gui-Bin Bian, Zeng-Guang Hou, Jun Zhao, Guodong Su, Hao Zhou, Liang Peng, Weiqun Wang, Simultaneous recognition and assessment of post-stroke hemiparetic gait by fusing kinematic, kinetic, and electrophysiological data, IEEE Trans. Neural Syst. Rehabil. Eng. 26 (4) (2018) 856–864.
[127] Shinan Zou, Jianbo Xiong, Chao Fan, Chuanfu Shen, Shiqi Yu, Jin Tang, A multi-stage adaptive feature fusion neural network for multimodal gait recognition, IEEE Trans. Biom. Behav. Identity Sci. (2023) 1–10.
[128] John Archila, Antoine Manzanera, Fabio Martínez, A multimodal Parkinson quantification by fusing eye and gait motion patterns, using covariance descriptors, from non-invasive computer vision, Comput. Methods Programs Biomed. 215 (2022) 106607.
[129] João Loureiro, Paulo Lobato Correia, Using a skeleton gait energy image for pathological gait classification, in: 2020 15th IEEE International Conference on Automatic Face and Gesture Recognition, FG 2020, IEEE, 2020, pp. 503–507.
[130] Yuxiang Feng, Jiabin Yuan, Lili Fan, Gaitfusion: Exploring the fusion of silhouettes and optical flow for gait recognition, in: International Conference on Artificial Neural Networks, Springer, 2023, pp. 88–99.
[131] Yingchan Cao, Ming Guo, Jianqiang Sun, Xiangyong Chen, Jianlong Qiu, Fall detection based on LCNN and fusion model of weights using human skeleton and optical flow, Signal Image Video Process. 18 (1) (2024) 833–841.
[132] Gulbadan Sikander, Shahzad Anwar, Ghassan Husnain, Rajermani Thinakaran, Sangsoon Lim, An adaptive snake based shadow segmentation for robust driver fatigue detection: A 3D facial feature based photometric stereo perspective, IEEE Access 11 (2023) 99178–99188.
[133] Jan Stenum, Cristina Rossi, Ryan T. Roemmich, Two-dimensional video-based analysis of human gait using pose estimation, PLoS Comput. Biol. 17 (4) (2021) e1008935.
[134] Andrea Bandini, Sia Rezaei, Diego L. Guarín, Madhura Kulkarni, Derrick Lim, Mark I. Boulos, Lorne Zinman, Yana Yunusova, Babak Taati, A new dataset for facial motion analysis in individuals with neurological disorders, IEEE J. Biomed. Heal. Inform. 25 (4) (2020) 1111–1119.
[135] Rahm Ranjan, David Ahmedt-Aristizabal, Mohammad Ali Armin, Juno Kim, Computer vision for clinical gait analysis: A gait abnormality video dataset, IEEE Access 13 (2025) 45321–45339.
[136] Arnav Mishra, Aditi Shetkar, Ganesh M. Bapat, Rajdeep Ojha, Tanmay Tulsidas Verlekar, XAI-based gait analysis of patients walking with Knee-Ankle-Foot orthosis using video cameras, 2024, arXiv preprint arXiv:2402.16175.
[137] Vanessa Buhrmester, David Münch, Michael Arens, Analysis of explainers of black box deep neural networks for computer vision: A survey, Mach. Learn. Knowl. Extr. 3 (4) (2021) 966–989.
[138] Zhan Zhang, Delong Jiang, Qingyan Wang, Pengqin Wang, Construction of an interpretable machine learning-based prediction model for the clinical effect on ischemic stroke in treatment with eye acupuncture combined with rehabilitation therapy, Zhongguo Zhen Jiu=Chin. Acupunct. Moxibustion 45 (5) (2025) 559–567.
[139] Hao Jin, Yan Luo, Peilong Li, Jomol Mathew, A review of secure and privacy-preserving medical data sharing, IEEE Access 7 (2019) 61656–61669.
[140] Lars Mündermann, Stefano Corazza, Thomas P. Andriacchi, The evolution of methods for the capture of human movement leading to markerless motion capture for biomechanical applications, J. Neuroeng. Rehabil. 3 (1) (2006) 6.
[141] Javier Ortells, María Trinidad Herrero-Ezquerro, Ramón A. Mollineda, Vision-based gait impairment analysis for aided diagnosis, Med. Biol. Eng. Comput. 56 (9) (2018) 1553–1564.
[142] Prachi Punyani, Rashmi Gupta, Ashwani Kumar, Human age-estimation system based on double-level feature fusion of face and gait images, Int. J. Image Data Fusion 9 (3) (2018) 222–236.